



Name: Cohort/Batch: Date: Score:

POWER BI PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A BI & Visualization

TOTAL: 10 QUESTIONS

Q1 What is Power BI and what problem in BI & Visualization does it solve?

Q6 How does Power BI handle reliability and high availability?

Q2 What are the core components or concepts in Power BI architecture?

Q7 What observability does Power BI provide out of the box?

Q3 How does Power BI compare to its main alternatives?

Q8 What are common performance bottlenecks in Power BI?

Q4 How do you get started with Power BI for a new project?

Q9 What security best practices apply when running Power BI?

Q5 What configuration options most affect Power BI's behavior in production?

Q10 What mistakes do teams commonly make when adopting Power BI?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Power BI is a tool or platform

Mitigation

Power BI is a tool or platform that automates or simplifies a specific concern in the BI & Visualization domain, reducing manual effort and operational complexity for engineering teams.

A6 Power BI provides HA through leader election,

Observability

Power BI provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Power BI has key building blocks that

Primitives

Power BI has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Power BI exposes metrics (Prometheus endpoint or

Automation

Power BI exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Power BI trades off simplicity against feature

Hardening

Power BI trades off simplicity against feature richness differently than its alternatives. Choose Power BI when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Power BI via its package manager

Gotchas

Install Power BI via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Power BI with the principle of

Assessment

Run Power BI with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency

Concurrency

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and

Documentation

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.